

Seven-Word Subcore Exclusions and a Six-Hole Plateau in the Ternary Covering Problem $K_3(7, 3)$

Ruturaj R Raval

Independent Researcher

ORCID: [0000-0003-4930-8981](https://orcid.org/0000-0003-4930-8981)

September 6, 2026

Abstract

Let $K_3(7, 3)$ be the minimum cardinality of a radius-three covering code in the ternary Hamming space $\{0, 1, 2\}^7$. The established global interval remains $11 \leq K_3(7, 3) \leq 12$. This report proves a scoped structural exclusion based on an explicit eight-word core C . For each $c \in C$, every set of at most four centers added to the seven-word subcore $C \setminus \{c\}$ leaves at least six words uncovered, and equality is attained. Consequently, no covering code of size at most eleven contains any of these eight subcores or an isometric copy. The order-18 stabilizer of C reduces the eight deletions to four representatives. Independent Rust and C++20 verifiers solve all four completion problems by exact pair reduction, building 9,500,440 pair entries and querying 347,365 eligible first pairs. The full eight-word core is also classified exactly: its three-center minimum is six, attained by eight triples in four stabilizer classes. Exact local replacement results and a direct diameter bound provide further structural constraints. The global interval is unchanged.

1 Introduction

A q -ary covering code is a set of words whose Hamming balls cover the entire ambient Hamming space. The quantity $K_q(n, R)$ denotes the minimum number of centers needed to cover $\{0, \dots, q-1\}^n$ with balls of radius R . Covering codes connect coding theory, extremal combinatorics, finite geometry, dominating-set problems, and exact optimization.

This report concerns the unresolved ternary value $K_3(7, 3)$. Kéri's covering-code table records

$$11 \leq K_3(7, 3) \leq 12$$

[1]. The associated change log dates the computer-derived lower bound $K_3(7, 3) \geq 11$ to October 22, 2008 [2]. Earlier ternary constructions and covering-code tables include the work of Hämäläinen and Rankinen [3] and of Östergård and Hämäläinen [4]. Recent semidefinite lower-bound techniques provide broader context but do not supersede the integer lower bound eleven in this cell [5].

The main result here is not a global bound improvement. Instead, it identifies eight explicit seven-word subcores that cannot occur inside any size-at-most-eleven cover, determines all optimal three-center completions of their common eight-word parent core, and describes an exact local replacement plateau around four resulting near-covers. The distinction between global and scoped statements is maintained throughout:

- The interval $11 \leq K_3(7, 3) \leq 12$ is a global fact.

- The subcore exclusion is complete for eight explicit seven-word sets and every isometric copy of each set.
- The full-core completion result is complete for one explicit eight-word set.
- The replacement results are complete for four specified near-cover classes.
- No classification of all eleven-word codes or all six-hole near-covers is claimed.

2 Definitions and notation

Let

$$\mathcal{H} = \{0, 1, 2\}^7.$$

For $x, y \in \mathcal{H}$, the Hamming distance is

$$d_H(x, y) = |\{i : x_i \neq y_i\}|.$$

For a nonempty set $D \subseteq \mathcal{H}$, write

$$d_H(x, D) = \min_{d \in D} d_H(x, d)$$

and define its radius-three uncovered set by

$$\mathcal{U}(D) = \{x \in \mathcal{H} : d_H(x, D) > 3\}.$$

Thus D is a radius-three covering code exactly when $\mathcal{U}(D) = \emptyset$.

The ambient space has

$$|\mathcal{H}| = 3^7 = 2187$$

words. Every radius-three Hamming ball has

$$\sum_{i=0}^3 \binom{7}{i} 2^i = 1 + 14 + 84 + 280 = 379$$

words.

An isometry of $\mathcal{H}$ is generated by a permutation of the seven coordinates and independent permutations of the three symbols in each coordinate. Such maps preserve Hamming distance, covering radius, uncovered-set cardinality, and code cardinality.

3 The core and its seven-word subcores

Consider the following eight-word set:

$$C = \{0000011, 0000102, 0000220, 0002121, \\ 1111022, 1111110, 1111201, 2222000\}.$$

Direct enumeration gives

$$|\mathcal{U}(C)| = 253.$$

For each $c \in C$, define

$$S_c = C \setminus \{c\}.$$

The stabilizer of C has order 18. Its action on the eight choices of c has four orbits, shown in Table 1.

Table 1: Deletion orbits and exact four-center completion results.

Representative c	Orbit size	$ \mathcal{U}(S_c) $	Pair threshold	Eligible first pairs	Minimum residual
0000011	3	370	183	86,319	6
0002121	1	358	177	81,290	6
1111022	3	396	196	82,464	6
2222000	1	518	257	97,292	6

Theorem 1 (Seven-word subcore exclusion). *For every $c \in C$ and every $T \subseteq \mathcal{H}$ with $|T| \leq 4$,*

$$|\mathcal{U}(S_c \cup T)| \geq 6.$$

Equality is attained for every c .

Proof. It is enough to solve one deletion from each of the four stabilizer orbits. Fix a representative c , let $h = |\mathcal{U}(S_c)|$, and suppose first that four added centers cover all but at most five of these h holes.

There are six unordered pairs among the four centers. Every covered hole belongs to at least three pair unions. Indeed, a hole covered by exactly one center belongs to the three pairs containing that center, and coverage by additional centers can only increase the number of pair unions containing it. Therefore the sum of the six pair-union coverage counts is at least

$$3(h - 5).$$

Some pair must cover at least

$$\left\lceil \frac{h - 5}{2} \right\rceil$$

holes.

Each representative has $2187 - 7 = 2180$ allowed added centers and

$$\binom{2180}{2} = 2,375,110$$

unordered candidate pairs. The verifier builds every pair-coverage mask, retains every first pair meeting the displayed averaging threshold, and queries all complementary pair masks for a union leaving at most five holes. The thresholds and retained first-pair counts are listed in Table 1. Across the four representatives, the calculation builds 9,500,440 pair entries and queries 347,365 eligible first pairs. No residual-at-most-five completion exists.

The Rust implementation permits overlapping pair witnesses and pads any found union to four distinct centers before accepting it. The independent C++20 implementation requires the complementary pairs to be disjoint and reconstructs the stabilizer separately. Both implementations reproduce all counts in Table 1.

For each representative c , the four additions

$$\{c, 1112212, 2220121, 2221212\}$$

leave exactly six holes, so the lower bound is attained. A set of fewer than four added centers can be augmented with distinct centers without increasing its residual. Stabilizer isometries transfer the result and the equality witnesses to every member of each deletion orbit. $\square$

Corollary 2 (Exclusion from size-at-most-eleven covers). *No radius-three covering code $D \subseteq \mathcal{H}$ with $|D| \leq 11$ contains any S_c . The same statement holds for every isometric copy of every S_c .*

Proof. If $S_c \subseteq D$ and $|D| \leq 11$, then $T = D \setminus S_c$ has at most four elements. Theorem 1 gives $|\mathcal{U}(D)| \geq 6$, contradicting that D is a cover. Isometries preserve all distances and cardinalities. $\square$

For each displayed representative, the equality witness in the proof leaves the same six words:

$$J = \{0011000, 0101000, 0110000, \\ 1001000, 1010000, 1100000\}.$$

Remark 3 (Scope). *Corollary 2 eliminates eight complete isometry families of candidate substructures. It does not eliminate an eleven-word cover containing none of these seven-word subcores.*

4 Exact completion of the eight-word core

There are $2187 - 8 = 2179$ possible added centers outside C .

Theorem 4 (Exact three-center completion of C). *For every $T \subseteq \mathcal{H} \setminus C$ with $|T| \leq 3$,*

$$|\mathcal{U}(C \cup T)| \geq 6.$$

Equality is attained. Among sets T of cardinality three, exactly eight unordered triples attain equality.

Proof. For each of the 2179 allowed centers, the verifier constructs its exact coverage mask on the 253 points of $\mathcal{U}(C)$. The direct Rust verifier enumerates every strictly ordered index triple, so each unordered triple is processed once. The number processed is

$$\binom{2179}{3} = 1,721,956,929.$$

For every triple it forms the union of the three coverage masks and counts the residual points. The minimum residual is six, and the second enumeration pass returns exactly eight distinct minimizing triples.

An independent Python verifier reconstructs $\mathcal{H}$, all radius-three balls, C , and $\mathcal{U}(C)$ separately. It groups centers only when their exact coverage masks on the current finite hole set are identical. The decision search and the complete minimizer expansion return the same minimum and the same eight unordered triples. Grouping equal masks changes only the number of repeated bit operations, not the set of candidate centers represented.

The exhaustive triple result also covers sets of fewer than three added centers. Any such set can be augmented to three distinct allowed centers. Adding centers can only decrease the uncovered set. Since every augmented triple still leaves at least six points, the original set leaves at least six points as well. $\square$

The eight minimizing triples are:

1112212	2220111	2221222
1112212	2220121	2221212
1112212	2220122	2221211
1112212	2220212	2221121
1112212	2220221	2221112
1121212	2212212	2220121
1211212	2122212	2220121
1222212	2111212	2220121.

Remark 5 (Additional classification). *Theorem 1 already excludes C , since every set containing C contains each of its seven-word subcores. Theorem 4 adds information not supplied by that exclusion: it determines all optimal three-center additions to the full core.*

5 A direct diameter bound

The subcore and full-core theorems are computational. The following independent result is a short counting argument.

Proposition 6 (Diameter at most four requires seventeen centers). *Every ternary length-seven code with covering radius at most three and diameter at most four has at least seventeen centers.*

Proof. Translate one code center to 0000000. Translation is an isometry, and the diameter assumption implies that every other center has Hamming weight at most four.

Consider the 128 full-support words

$$F = \{1, 2\}^7.$$

A center of weight at most three is at distance at least four from every word of F , so it covers none of them at radius three. A center of weight four covers a word of F only when the four nonzero symbols agree on the center's support. The remaining three symbols are free. Consequently, each weight-four center covers exactly $2^3 = 8$ words of F . The translated zero center covers none of F . Therefore

$$128 \leq 8(|D| - 1),$$

which gives $|D| \geq 17$. □

Corollary 7. *Every radius-three ternary length-seven cover with at most sixteen centers has diameter five, six, or seven.*

This statement is global: it applies to every code under its hypotheses. It does not by itself distinguish between sizes eleven and twelve.

6 Four six-hole near-cover classes

Each of the eight triples in Theorem 4 produces an eleven-word code with exactly six uncovered words. In the normalization used by the verifier, the common uncovered set is the set J displayed after Theorem 1. On the first four coordinates these are the binary weight-two words, so the induced geometry is the Johnson graph $J(4, 2)$; the final three coordinates are zero.

The stabilizer of C within the normalized hole stabilizer has order 18. Its action partitions the eight minimizing triples into four orbits. Representatives are shown in Table 2.

Exact code-hole canonicalization under coordinate and coordinatewise symbol permutations proves that these four representatives are mutually non-isometric. Equivalently, the four retained eleven-word near-covers belong to four distinct isometry classes. This is a statement about the retained and normalized configurations, not a proof that every six-hole eleven-word near-cover belongs to one of these classes.

Table 2: Representative three-word tails. Each row, joined with C , leaves exactly the six words in J uncovered.

Class		Tail T		
A	1112212	2220121	2221212	
B	1121212	2212212	2220121	
C	1112212	2220111	2221222	
D	1112212	2220212	2221121	

7 Exact replacement neighborhoods

For an eleven-word code D , a *strict one-center replacement* removes one center of D and adds one center from $\mathcal{H} \setminus D$. A *strict two-center replacement* removes two centers of D and adds two distinct centers from $\mathcal{H} \setminus D$. The replacement code is required to have eleven distinct centers.

For each class in Table 2, the Python verifier exhaustively determines:

- the minimum number of uncovered words after a strict one-center replacement;
- the minimum after a strict two-center replacement;
- every strict two-center replacement that preserves the six-hole residual; and
- the isometry class of every such six-hole neighbor.

The results are in Table 3.

Table 3: Exact strict replacement results. The final column gives the number of six-hole neighbor incidences by target class.

Class	Best one-center residual	Best two-center residual	Six-hole neighbors	Target classes
A	15	6	14	$B : 6, C : 6, D : 2$
B	15	6	6	$A : 2, B : 4$
C	18	6	5	$A : 2, C : 2, D : 1$
D	15	6	5	$A : 2, C : 3$

For each source class, there are

$$\binom{11}{2} \binom{2176}{2} = 130,152,000$$

raw strict two-center replacement candidates. Across the four source classes, the exact finite search therefore represents

$$520,608,000$$

raw candidates. Coverage-mask classes reduce repeated work while retaining every center pair represented by each mask pair.

The strict one-center minima show that no single strict replacement stays on the six-hole plateau. Strict two-center replacement can remain on the plateau, and the 30 source-neighbor incidences in Table 3 give its directed local adjacency. No strict one-center or strict two-center replacement of these four representatives produces fewer than six uncovered words.

8 Verification design

8.1 Independent geometry reconstruction

The Rust, C++20, and Python implementations generate all 2187 ternary words from integer encodings and compute Hamming distance directly. The two primary subcore verifiers separately reconstruct the eight-word core, its stabilizer, the deletion orbits, all four residual spaces, and every candidate-center coverage mask.

8.2 Exact Rust pair reduction

The Rust subcore verifier builds all 2,375,110 unordered pair unions for each representative. A binary mask tree stores subtree intersections and unions, allowing exact rejection or acceptance of a complementary-pair query without scanning every leaf. The averaging lemma determines the complete first-pair query set.

The Rust query permits pair overlap. If a complementary pair is found, the union is padded to four distinct centers and its actual residual is recomputed before acceptance. The test suite compares this complete pipeline with direct four-set enumeration on 6,480 deterministic synthetic instances.

8.3 Independent C++20 pair reduction

The C++20 verifier does not import Rust orbit or pair-index data. It reconstructs the order-18 stabilizer from Hamming isometries, builds a different pair tree, and queries only disjoint complementary pairs. Its finite controls perform 21,504 pair-index queries and 70 direct four-center checks. Additional boundary controls increase these totals to 27,984 pair-index queries and 140 direct four-center sets, including 63-, 64-, and 65-bit residual masks. It independently reproduces the exclusion result and all theorem-defining counts from the Rust implementation. The published residual-six equality witness is supplied to both implementations and verified directly rather than discovered independently.

8.4 Direct Rust triple scan

The separate full-core Rust verifier stores the 253 core holes in four machine-word blocks. It excludes the eight core centers, builds one exact hole-coverage mask for each allowed center, and executes three nested loops over strictly increasing candidate indices. It checks the expected combinatorial count before accepting the result. A second complete pass enumerates all triples attaining the computed minimum and verifies that there are exactly eight.

The scan is deterministic. Work may be partitioned among threads by the first triple index, but worker count does not alter the candidate set, ordering, minimum, or final sorted minimizer list. A single-worker run is included as the simplest replay.

8.5 Exact Python mask-class enumeration

The Python verifier represents subsets of $\mathcal{H}$ as integer bitmasks. For the core-completion problem, it groups candidate centers by their exact coverage mask on the current residual set. It first proves that no triple has residual at most five, then expands every mask-class combination at residual six to recover the complete set of center triples.

The same exact-mask principle is used for two-center replacement neighborhoods. The test suite compares mask-pair pruning with direct enumeration on all 65,536 four-mask synthetic instances and checks the three-center decision routine against independent exhaustive controls.

In continuous integration, the one-worker Rust output, parallel Rust output, and C++20 output are parsed and compared field by field with the checked machine-readable result record. This binds the published orbit members, counts, drift checks, minima, and witnesses to fresh verifier executions.

8.6 CNF availability and claim boundary

A deterministic CNF encoding is available for the fixed-core completion problem. It contains primary variables for added centers, slack variables for uncovered core holes, an at-most-three center constraint, and an at-most- r residual-hole constraint.

No unsatisfiability result or checked proof for the $r = 5$ CNF is claimed in this report. The CNF is an auxiliary encoding for future cross-checking. Theorem 4 rests on the two finite exhaustive full-core verifiers described above, while Theorem 1 rests on the independent Rust and C++20 pair reductions.

9 Reproducibility

From the repository root, the central result is replayed with:

```
make verify-result
```

The two primary verification paths can be run separately:

```
SEVEN_CORE_DIRECT_WORKERS=1 \
  cargo run --locked --release --manifest-path search/Cargo.toml \
  --bin seven_core_direct
```

```
make build/seven-core-independent
build/seven-core-independent --workers 1
```

The supporting full-core and plateau computations are replayed with:

```
make core-direct-single
make core-direct
make plateau
```

The complete test and lint commands are:

```
make verify
make test
make test-search
make lint-search
```

The optional CNF can be generated without making a solver claim:

```
make core-cnf
```

The retained near-covers, expected exact counts, source-file hashes, and canonical class data are recorded in:


```

data/search/seed_101_best_11.txt
data/search/seed_202_best_11.txt
data/search/seed_303_best_11.txt
data/search/seed_404_best_11.txt
data/search/plateau_classes.json
evidence/seven-core-results.json

```

10 Significance

The principal contribution is a complete structural exclusion inside a long-unresolved one-unit global interval. Its significance is fourfold.

1. A pattern first encountered through construction search is converted into a finite theorem: none of eight explicit seven-word subcores can occur in any size-at-most-eleven cover, even after arbitrary choice of the remaining four centers.
2. The pair-averaging reduction gives a reusable exact method for four-center completion questions over finite residual spaces.
3. The six-hole behavior is resolved exactly for the full core. The minimum residual, all eight optimal triples, their four stabilizer classes, and the local one-center and two-center replacement structure are reproducible.
4. Proposition 6 removes all diameter-at-most-four candidates from every search for small covers, leaving only diameters five, six, and seven for codes of size at most sixteen.

These results can be used to prune exact-cover or SAT searches, construct regression benchmarks for independent solvers, identify forbidden substructures in candidate eleven-word codes, and guide larger neighborhood moves away from the observed six-hole plateau.

11 Limitations and open directions

The global interval remains

$$11 \leq K_3(7, 3) \leq 12.$$

In particular:

- No eleven-word covering code is constructed.
- No exclusion of all eleven-word covering codes is proved.
- No new global lower or upper bound is claimed.
- The historical global lower bound eleven is not independently certified here.
- The excluded seven-word subcores are not claimed to occur in every hypothetical eleven-word cover.
- The four retained six-hole classes are not claimed to classify all eleven-word near-covers.
- The CNF encoding has no claimed $r = 5$ unsatisfiability proof.

- No compact proof certificate independent of the two verifier implementations is supplied.
- Repeated search convergence and solver timeouts are not used as mathematical evidence.

Natural next steps are to derive additional forbidden subcores, import all such exclusions into the diameter-five through diameter-seven branches, build a global canonical classification of candidate eleven-word structures, and either construct an eleven-word cover or produce a complete checked exclusion certificate.

12 Code and data availability

The versioned repository release associated with this report contains the independent Rust and C++20 subcore verifiers, the Python and Rust full-core verifiers, exact tests, retained near-cover representatives, the machine-readable result record, the canonical plateau manifest, the optional CNF generator, and all replay commands listed above. The separate deterministic manuscript bundle contains only the files needed to compile this report. The repository release and its archival record should be cited together so that the source revision and generated results remain fixed. The paper-inclusive archival release is version v0.2.1, DOI <https://doi.org/10.5281/zenodo.22647770>, under concept DOI <https://doi.org/10.5281/zenodo.22452732>. This archival and documentation patch adds an explicit compiled PDF and deterministic source bundle; the theorem, proof, certificates, retained data, and computations are unchanged.

Acknowledgment of computational scope

All central claims are finite statements over the explicitly defined space $\mathcal{H}$. The report separates direct mathematical arguments, complete finite computations, optional encodings, and unresolved global questions. No inference is drawn from an incomplete solver run.

References

- [1] G. Kéri, *Tables for ternary covering codes*, https://old.sztaki.hu/~keri/codes/3_tables.pdf, accessed September 6, 2026.
- [2] G. Kéri, *Covering-code table change log*, entry dated October 22, 2008, <https://old.sztaki.hu/~keri/codes/index.htm>, accessed September 6, 2026.
- [3] H. Härmäläinen and S. Rankinen, “Upper bounds for football pool problems and mixed covering codes,” *Journal of Combinatorial Theory, Series A*, vol. 56, no. 1, pp. 84–95, 1991. doi:10.1016/0097-3165(91)90024-B.
- [4] P. R. J. Östergård and H. O. Härmäläinen, “A new table of binary/ternary mixed covering codes,” *Designs, Codes and Cryptography*, vol. 11, no. 2, pp. 151–178, 1997. doi:10.1023/A:1008228721072.
- [5] D. Gijswijt and S. Polak, “Semidefinite lower bounds for covering codes,” arXiv:2504.01932v2, 2026. <https://arxiv.org/abs/2504.01932v2>.